CLASS : B. E. (E &TC) **SUBJECT : ESD**

EXPT. NO. : 07 **DATE :**

Aim:

Write Analog-to-Digital Converter (ADC) drivers using HAL functions.

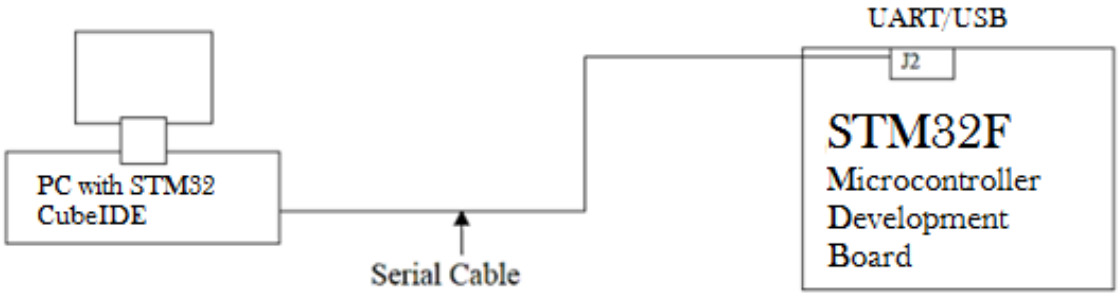
Objectives:

Create project and write code using HAL function to read analog input using inbuilt ADC of STM32F microcontroller.

Requirements:

STM32FXX Microcontroller Development Board, Personal Computer, STM32CubeIDE Software, Connecting wires, Power Supply, etc.

Block Diagram:



Theory:

Analog-To-Digital Converters (ADC):

An ADC (Analog-To-Digital) converter is an electronic circuit that takes in an analog voltage as input and converts it into digital data, a value that represents the voltage level in binary code. The ADC samples the analog input whenever you trigger it to start conversion. And it performs a process called quantization so as to decide on the voltage level and its binary code that gets pushed in the output register.

STM32 ADC:

The STM32F103C8 (Blue Pill) & STM32F432KC have a 12-bit ADC which is a successive approximation analog-to-digital converter. It has up to 18 multiplexed channels allowing it to measure signals from sixteen external and two internal sources. A/D conversion of the various channels can be performed in single, continuous, scan, or discontinuous mode. The result of the ADC is stored in a left-aligned or right-aligned 16-

bit data register.

The analog watchdog feature allows the application to detect if the input voltage goes outside the user-defined high or low thresholds. The ADC input clock is generated from the PCLK2 clock divided by a Prescaler and it must not exceed 14 MHz.

ADC Features:

- 12-bit resolution
- Interrupt generation at End of Conversion, End of Injected conversion and Analog watchdog event
- Single and continuous conversion modes
- Scan mode for automatic conversion of channel 0 to channel ‘n’
- Self-calibration
- Data alignment with in-built data coherency
- Channel by channel programmable sampling time
- External trigger option for both regular and injected conversion
- Discontinuous mode
- Dual-mode (on devices with 2 ADCs or more)
- ADC conversion time: 1 µs at 56 MHz (1.17 µs at 72 MHz)
- ADC supply requirement: 2.4 V to 3.6 V
- ADC input range: VREF– ≤ VIN ≤ VREF+
- DMA request generation during regular channel conversion

STM32 ADC Resolution, Reference, Formulas:

- STM32 ADC Resolution:- The STM32 ADC has a resolution of 12-Bit which results in a total conversion time of SamplingTime+12.5 clock cycles.
- ADC Reference Voltage:- The ADC reference voltage pins are defined in the datasheet and assumed to be connected to a voltage level in a certain range.
- STM32 ADC Formulas:
    - ADC Conversion Time

      Tconv = Sampling time + 12.5 cycles
    - ADC Sampling Rate

      SamplingRate = 1 / Tconv

- ADC Result Voltage (Analog Input Value)

  Vin = ADC_Res x (Reference Voltage / 4096)

  Where Reference Voltage = (VREF+) – (VREF-)

Different Ways To Read STM32 ADC:

1. The Polling Method
2. The Interrupt Method
3. The DMA Method

Building the Project:

Follow the debug and execution procedure mentioned in the previous experiment.

Conclusion: _____

_____

_____